TECHNICLE REPORT WRITING FOR CA2

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STREAM- COMPUTER SCIENCE AND BUSINESS SYSTEM

SUBJECT NAME- OBJECT ORIENTED PROGRAMING

YEAR- SECOND (FOURTH SEMESTER)

REGISTRATION NO- 221870110094

SUBJECT CODE- PCC CSBS-402

ROLL NUMBER- 18731122015

SESSION- 2022-2026

TOPIC: ‘C++’ INPUT AND OUTPUT OPERATION IN RESPECT OF ‘C’ LANGUAGE

INTRODUCTION:

❖ HISTORY OF 'C++':

Bjarne Stroustrup, a researcher, started working at AT&T Bell Laboratories in 1979.

As computers became more sophisticated and powerful in the 1970s, limitations in the C programming language began to appear.

Two factors drove Bjarne Stroustrup's creation of a new language in the early 1980s:

- The Object-Oriented Programming features of Simula, a language other than C that provided an alternative method of programming. Classes would allow for the abstraction, organization, and representation of any type of code.

- C is a systems programming language that provided the capability to perform complex low-level computing operations and reach very near to the machine hardware.
  Higher level abstraction was made possible by combining these two concepts without sacrificing C's low-level performance. So, the language '*C with classes*' was created.

USES:
Once more, C++ gives the programmer intimate access to and control over the computer while offering large-scale systems efficiency and excellent performance. In addition, it is sufficiently abstracted from the machine to be higher level.

Web browsers including Google Chrome, Mozilla Firefox, and Safari, gaming engines, computer graphics and apps, virtual reality applications, and web browser extensions are all frequently made in C++. Additionally, the Google Search Engine is C++-based.

❖ HISTORY OF 'C':

Dennis Ritchie began by enhancing the B programming language and ultimately produced the C programming language.

Programs created in C can be transferred and utilized on a range of machine architectures since it is a portable language. Programmers may access low level functionalities with its direct mapping to machine code, which makes it incredibly quick and simple to compile. In 1972, they rewrote the Unix operating system in C.

- In 1977, Dennis Ritchie and Brian Kernighan co-wrote the book "C programming language," which established guidelines for using the language. The phrase gained considerably greater traction after this work.

- The development of C led to the birth of numerous additional programming languages, making it incredibly important in the history of computers. It is frequently referred to as the "mother" of all programming languages because of this.

USES:

For really difficult, low-level computing jobs where speed, effectiveness, and close access to the machine are essential, C is frequently utilized.

C provides programmers freedom and thinks they know what they're doing.
Because of this, it is the preferred language for developing operating systems, embedded systems,

systems programming, kernels and drivers, compilers, and the expanding field of Internet of Things (IoT) applications.

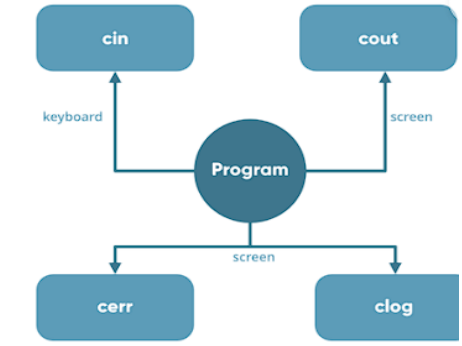
As a result, C++ is a superset of C, which indicates that it was built upon and an extension of C. All C++ does is give the C language more features.

METHODOLOGY:

BASIC INPUT OUTPUT OPERATION IN 'C++':

Libraries for C++ are included with the program, giving us numerous options for input and output. Input and output in C++ take the form of a stream, which is just a collection of bytes.

Input Stream: This process is referred to as input if the bytes are flowing from the device—such as a keyboard—to the main memory.
Output Stream: This process is referred to as output if the bytes are flowing in the other direction, that is, from main memory to a device (display screen).



The following header files are accessible in C++ for input/output operations:

- The acronym for standard input-output stream is iostream. Definitions for objects like Cin, cout, cerr, and so forth are contained in this header file.
- The acronym for input-output manipulators is iomanip. These files include declarations for methods that are used to manipulate streams. Definitions for setw, set precision, etc., are contained in this file.
- fstream: The file stream is mostly described in this header file. The handling of data being written into a file as output or received from a file as input is done by this header file.
- bits/stdc++: All of the standard libraries are contained in this header file. Using this file during programming contests is a fantastic idea if you want to cut down on the amount of time you spend on tasks, especially if your rank is time-sensitive.

In C++, "using namespace std;" is frequently used after the header files. The namespace std contains all of the definitions for the standard library, which explains why. Since the library functions are not specified at the global scope, namespace std is used to access them. In order to avoid writing STD: at the beginning of every line (e.g., STD: cout etc.). Consult this article to learn more.

The two instances of the iostream class, cout in C++ and Cin in C++, are frequently used for printing outputs and receiving inputs, respectively. In C++, these are the two simplest ways to receive input and output it. In order to utilize Cin and cout in C++, the program needs to include the iostream header file.

CODE:

```
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main ()
{
    char sample [] = "HELLO";
    cout << sample << " - WELCOME TO CODE WORLD ";
    return 0;
}
```

OUTPUT:

HELLO – WELCOME TO CODE WORLD

Time Complexity: O (1)
Space Complexity: O (1)

The insertion operator (\\) in the program above inserts the string "A computer science portal for geeks" after the value of the string variable sample, and the result is displayed on the screen using the standard output stream cout.

standard input stream (Cin): A computer's keyboard is typically its input device. The C++ Cin statement reads input from the standard input device, which is typically a keyboard. It is an instance of the class iostream.
To read inputs, the object Cin is used in conjunction with the extraction operator (>>). The data entered using the keyboard in object Cin is extracted by the extraction operator.

CODE:

```
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main ()
{
int age;
    cout << "Enter your age:";
    cin >> age;
    cout << "\nYour age is: " << age;
    return 0;
}
```

INPUT: 18

OUTPUT: `Enter your age:`
`Your age is: 18`

Time Complexity: O(1)
Space Complexity: O(1)

The user is prompted to enter their age in the program above. The input device is linked to the object Cin. Using the extraction operator (>>), the user's age is taken out of Cin and put into the variable age that is located on the right side of the extraction operator.

- Un-buffered standard error stream (cerr): The errors are output using the C++ cerr standard error stream. This is another example of an iostream class instance. Since cerr in C++ is not buffered, it is utilized in situations where the error message must be shown right away. It lacks a buffer to hold the error message and show it at a later time.
- The primary distinction between cerr and cout is that if you use "cerr," the error is not recorded in the file and is instead diverted to the output when you use "cout" to redirect output. (This is the meaning of un-buffered. It is unable to store the text.

CODE:

```
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main ()
{
    cerr << "An error occurred";
    return 0;
}
```

OUTPUT: `An error occurred`

Time Complexity: O (1)
Space Auxiliary: O (1)

Similar to cerr, the buffered standard error stream (clog) is an instance of the ostream class that is used to display errors. However, instead of being displayed immediately, the error is first entered into a buffer and is kept there until it is not completely filled. or flush () is not used to explicitly flush the buffer. Additionally, the error message will appear on the screen.

CODE:

```
#include <iostream>
using namespace std;
int main ()
{
    clog << "An error occurred";
    return 0;
}
```

OUTPUT:

```
An error occurred
```

Time Complexity: O (1)
Space Complexity: O (1)

❖ CONCLUSION:

In conclusion, there are a number of important ways in which C++ input and output procedures are superior than the C language paradigm. The switch from functions to C++'s stream-based method (such as std:c out and std:c in) provides a more approachable and type-safe method of managing activities involving input and output.
Runtime error probability is decreased by type mismatch detection at compile time, which is made possible by C++'s type safety. Furthermore, input/output functions can be connected with classes and objects thanks to C++'s object-oriented characteristics, offering a more structured and adaptable method. Additionally, C++'s robust standard library support provides high-level abstractions for input/output operations, simplifying the handling of files, strings, and other data sources. This is in contrast to C, which for comparable tasks depends on additional libraries or platform-specific functions.
Overall, C++ input and output operations add a great deal of usefulness, safety, and extensibility while building upon the foundation made by C. Compared to the conventional C methodology, C++ offers a more contemporary and effective way to handle input and output duties by utilizing streams, type safety, object-oriented programming, and a robust standard library.

❖ REFERENCES:

(PDF) Input/Output in C++ (researchgate.net)
inputoutput.pdf (csulb.edu)
Programming and Problem Solving with C++, 2/e (mtsu.edu)